

Copilot Studio Training – Day 05

Agenda:

- Use Dataverse as Knowledge Source
- Use Azure SQL as Knowledge Source
- Use Azure AI Search as Knowledge Source

Use Case 01: SalesGenie with Dataverse as a Knowledge Source

Overview

In this use case, we will implement the capabilities of **SalesGenie** by integrating an knowledge source: a Dataverse table named "**Sales**." This table contains valuable sales data that will empower SalesGenie to deliver deeper insights and more tailored recommendations. By connecting this Dataverse table, SalesGenie can analyze sales information dynamically and respond to user queries with precision.

Below is the **Dataverse Schema** for the "Sales" table, which will serve as the foundation for this integration. This schema ensures that SalesGenie can access structured, high-quality data to provide actionable insights and improve sales outcomes.

Copilot Studio - Class 04 > Tables > Sales > Columns ▾

Display name ↑ ▾		Name ▾ ▾	Data type ▾
Business Area	⋮	ppt_BusinessArea	Single line of te:
Customer Active or Inactive	⋮	ppt_CustomerAc...	Yes/no
Customer Name Primary name color	⋮	ppt_CustomerNa...	Single line of te:
Profit Amount	⋮	ppt_ProfitAmount	Currency
Profit Amount (Base)	⋮	ppt_profitamoun...	Currency
Region	⋮	ppt_Region	Single line of te:
Sales	⋮	ppt_SalesId	Unique identifie
Sales Amount	⋮	ppt_SalesAmount	Currency
Sales Amount (Base)	⋮	ppt_salesamount...	Currency
Sales Channel	⋮	ppt_SalesChannel	Choice
Year	⋮	ppt_Year	Whole number

Solution:**Prerequisite (Create a Table):**

If you haven't created a Dataverse table by your own, then you can use the below prompt to create a table.

Prompt:

I want to create one Sales Table which shows some Customer Name, Customer ID, Business Area, Region, Sales Amount, Profit Amount, Year, Customer Active or Inactive etc. You can suggest me some other columns as well.

Connect Dataverse as Data source.

Use Below suggested synonyms and description for each column:

Column Name	Description	Data Type	Synonyms
Business Area	Represents the specific area or department responsible for the business transaction.	Single line of text	Division, Department, Sector, Functional Area
Customer Active or Inactive	Indicates whether the customer is currently active or inactive.	Yes/No	Customer Status, Account Status, Active Flag
Customer Name	Stores the name of the customer	Single line of text	Client Name, Account Name, Buyer Name
Profit Amount	The total profit earned from a specific sale or transaction.	Currency	Net Profit, Earnings, Gain, Margin
Region	The geographical location or region associated with the sale.	Single line of text	Territory, Zone, Area, Location
Sales	A unique identifier for each sale record.	Unique identifier	Transaction ID, Deal ID, Sales Record ID
Sales Amount	The total amount generated from a specific sale.	Currency	Revenue, Total Sale, Gross Sales, Invoice Value
Sales Channel	The channel through which the sale was made (e.g., Online, Retail).	Choice	Distribution Channel, Selling Platform, Medium
Year	The year in which the sale occurred.	Whole number	Fiscal Year, Calendar Year, Reporting Year

Try to add below Glossary for each column to get proper Result:

Glossary for SalesGenie Dataverse Table

- **Business Area:** A functional division or department within the organization responsible for business transactions.
- **Customer Active or Inactive:** A status indicator showing whether the customer account is currently engaged with the business (Active) or dormant (Inactive).
- **Customer Name:** The full name or identifier for the customer or client. Used for tracking and managing sales relationships.
- **Profit Amount:** The net profit earned from a sale after subtracting costs. This represents the financial gain from the transaction.
- **Region:** A geographical location associated with the sale, such as North, South, East and West.

- **Sales Amount:** The total revenue generated from a sale before subtracting any costs or expenses.
- **Sales Channel:** The medium or platform through which a sale was made, such as Online, Retail, or Wholesale.
- **Year:** The calendar or fiscal year in which the sale occurred.

Questions to Test:

1. **Customer Activity:**

- a. How many customers are active in the Retail category?
- b. How many inactive customers exist in the Electronics category?
- c. List all active customers in the North region.

2. **Sales Analysis:**

- a. What is the total sales amount for the top 3 regions?
- b. What are the top 3 sales amounts for the North region?
- c. Which year generated the highest sales for the Retail category?

3. **Profit Insights:**

- a. What is the highest profit earned for the Retail category?
- b. Which customer contributed to the highest profit in the Electronics category?
- c. Show the top 5 sales records with the highest profit amount in 2024.

4. **Inactive Customer Insights:**

- a. How many customers were inactive in 2025?

5. **Region and Category Trends:**

- a. Which region generated the highest profit in 2023?
- b. What is the total profit amount for the Clothing category in the South region?
- c. Which sales channel performed best for the Retail category this year?

6. **Year-on-Year Comparisons:**

- a. Compare sales in 2024 and 2025 for the Retail category.
- b. How has the profit for the North region changed over the past 3 years?

7. How many customers are active for Clothing Category?
8. North Region Sales amount by top 3
9. Highest Profit for Jewelry category
10. For which customer did you get that profit?
11. Inactive customers in 2025

Number of Tables

- **Dataverse Knowledge Sources:** In classic orchestration mode, you can add up to 15 tables per knowledge source

References:

<https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-add-dataverse>

CONFIDENTIAL

Use Case 02 – Enhancing LeadGenie with Azure SQL as Knowledge Base

Objective

LeadGenie is an AI-powered chatbot designed to streamline and automate the process of managing sales leads within your organization. By seamlessly integrating with your Azure SQL database, LeadGenie enables your sales team to quickly view, update, and track sales leads, ensuring that no lead falls through the cracks.

Knowledge Source: Azure SQL

Solution

Agent Name: LeadGenie

Description:

LeadGenie is an AI-powered chatbot designed to streamline and automate the process of managing sales leads within your organization.

Instructions:

- **Professional but Friendly:** Maintain a balance between business professionalism and casual helpfulness.
- **Responsive and Efficient:** The agent should prioritize delivering information quickly and concisely.
- **Supportive and Motivational:** Encouraging users to stay engaged with their sales activities and providing gentle reminders for follow-ups.

Prerequisite (Create a Table):

If you haven't created a Azure SQL table by your own, then you can use it with the scripts I have provided here.

Connect Dataverse as Data source.

Use Below suggested synonyms and description for each column:

Here's the table structure in a detailed format with column names, data types, descriptions, and synonyms:

Column Name	Data Type	Description	Synonyms
LeadID	NVARCHAR(50)	Unique identifier for each lead.	ID, Identifier, UniqueID
LeadName	NVARCHAR(255)	Name of the lead.	Name, ContactName
Email	NVARCHAR(255)	Email address of the lead.	EmailAddress, ContactEmail
Phone	NVARCHAR(50)	Phone number of the lead.	PhoneNumber, ContactPhone
CompanyName	NVARCHAR(255)	Company associated with the lead.	BusinessName, OrganizationName
Status	NVARCHAR(50)	Status of the lead (New, In Progress, etc.).	LeadStatus, CurrentStatus
LeadSource	NVARCHAR(50)	Source of the lead (Website, Referral, etc.).	Source, Origin
AssignedTo	NVARCHAR(255)	Sales representative assigned to the lead.	AssignedRep, SalesRep
Priority	NVARCHAR(50)	Priority level of the lead (High, Medium, Low).	ImportanceLevel, PriorityLevel
LastContacted	DATETIME	Date the lead was last contacted.	LastFollowUp, LastTouchPoint
NextFollowUpDate	DATETIME	Scheduled date for the next follow-up.	FollowUpDate, NextTouchPoint
Notes	NVARCHAR(MAX)	Additional notes about the lead.	Comments, Remarks, History
EstimatedValue	DECIMAL(18, 2)	Potential revenue value from the lead.	PotentialRevenue, DealValue
CreatedDate	DATETIME	Date the lead was created.	CreationDate, EntryDate
ModifiedDate	DATETIME	Date the lead was last modified.	UpdateDate, LastModified

Questions to Test

1. Can you summarize notes for Gamma Enterprises?
2. How many Leads coming via Referral as LeadSource?
3. Total Leads generated this year
4. Total Leads generated this 2024
5. Can you just give me the summary of each company whose LeadSource is Referral

Azure AI Search + Copilot Studio

Why Use Azure AI Search Instead of Uploading Files Directly?

- ◇ Direct Upload (Knowledge Tab)
- ✓ Easy
- ✓ Quick
- ✗ No control over indexing
- ✗ No control over embeddings model
- ✗ Limited customization

Azure AI Search (Recommended for Production)

- ✓ Full control over indexing
- ✓ Choose embeddings model
- ✓ Add custom fields & labels
- ✓ Better ranking & search quality
- ✓ Semantic search support
- ✓ Scalable for enterprise

Key Takeaway:

Azure AI Search gives significantly better results and flexibility compared to direct file upload.

Architecture Overview (RAG Pattern)

Blob Storage (Document)



Azure AI Search (Index + Vector Search)



Embeddings Model (Azure OpenAI)



Copilot Studio Agent



User Question → Grounded Answer

This is a **Retrieval-Augmented Generation (RAG)** implementation.

Required Azure Resources (3 Total)

Resource	Purpose
Azure AI Search	Creates index & performs vector search
Azure Blob Storage	Stores documents
Azure OpenAI / AI Foundry	Deploy embeddings model

Step-by-Step Implementation

STEP 1 – Create Azure AI Search

Search: **AI Search** → Create

Important Settings:

- Resource Group: Keep all resources in one group
- Pricing Tier:
 - Free (50MB only)
 - Basic (Recommended)
 - Standard (Advanced features)

Why Basic or Standard?

- ✓ More storage
- ✓ More vector quota
- ✓ Semantic ranking
- ✓ Better performance

STEP 2 – Create Storage Account

Search: **Storage Account** → Create

Key Settings:

- Same Resource Group
- Performance: Standard
- Primary Service: Blob Storage

After creation:

Create Container

Example:

cs-human-resources

Upload:

- Adani HR Employee Handbook (PDF)

STEP 3 – Create Azure OpenAI Resource

Search: **Azure OpenAI** → Create

After deployment:

Go to Azure AI Foundry → Deploy Model

Deploy an **Embeddings Model**

What is an Embeddings Model?

Embeddings model:

- ✓ Converts text into vectors (numbers)
- ✓ Enables semantic similarity search
- ✓ Allows vector-based ranking

Example models:

Model	Quality
text-embedding-3-large	Best
text-embedding-3-small	Basic
ada-002	Older

Create Index in Azure AI Search

Go to:

AI Search → **Import Data(New)**

Configuration:

1. Data Source → Azure Blob Storage
2. Processing Type → RAG
3. Select:
 - Storage Account
 - Container
4. Select:
 - Azure OpenAI Resource
 - Embeddings Model Deployment
5. Authentication → API Key
6. Schedule → Run Once (or scheduled for production)

Click **Create**

What Gets Created?

Azure automatically creates:

Component	Purpose
Index	Stores searchable data
Indexer	Processes files into index

If error occurs:

1. Go to Indexer
2. Check error
3. Click Reset
4. Run Again

Testing Search in Azure

Use "Search Explorer" in Azure AI Search.

Connect Azure AI Search to Copilot Studio

In Copilot Studio:

1. Go to Agent
2. Click Add Knowledge
3. Select Azure AI Search
4. Create New Connection

You need:

Field	From Azure
Endpoint URL	AI Search Overview
Admin Key	AI Search → Keys

After connection:

- Select Index
- Click Add to Agent
- Status becomes "Ready" instantly

⚡ Why Instant?

Because indexing already happened in Azure.

Important Concepts Explained

◆ Index

Searchable database of document chunks.

◆ Indexer

Reads files from blob storage and pushes them into index.

◆ Vectorization

Converting text into numerical representations.

Example:

Maternity Leave → [0.234, 0.876, -0.332, ...]

Enables semantic similarity search.

◆ RAG (Retrieval Augmented Generation)

Instead of LLM guessing:

1. Retrieve relevant document parts
 2. Send them to LLM
 3. Generate grounded answer
-

Why Azure AI Search Gives Better Results

Compared to Upload Method:

Upload Method	Azure AI Search
Limited control	Full control
Fixed embeddings	Choose embeddings
Basic ranking	Semantic ranking
Slower scaling	Enterprise scalable
No enrichment	AI enrichment support

Production Best Practices

- ✓ Use Basic or Standard tier
 - ✓ Use text-embedding-3-large
 - ✓ Enable semantic ranking
 - ✓ Schedule index refresh
 - ✓ Organize documents by containers
 - ✓ Monitor indexer logs
-

When to Use Azure AI Search?

Use it when:

- Large document sets
- Enterprise chatbot
- HR/Legal/Policy agents
- Need high accuracy

- Need vector search
- Need advanced filtering

Use Case 03: “Policy & HR Helpdesk Agent” using Azure AI Search (RAG)

1) Business Problem

A company has multiple internal documents like:

- Working Hours Guidelines
- Employee Grievance Policy (SPEAK UP)
- Code of Conduct

Employees keep asking repetitive questions:

- “How many hours can I work per week?”
 - “How do I raise a grievance and what’s the timeline?”
 - “What’s considered harassment / conflict of interest?”
- HR teams waste time answering the same queries, and employees often get inconsistent answers.

2) Goal of the Agent

Build an internal **HR Policy Assistant Agent** that:

- Answers employee questions **only from official policy documents**
- Provides **source-cited** responses (document + section/page)
- Gives step-by-step guidance for processes (like “how to raise grievance”)
- Escalates to HR when the question is outside policy scope

3) Solution Overview (Architecture)

Azure AI Search + Agent (RAG)

Data sources

- PDFs / Word documents stored in SharePoint / Blob Storage

Indexing

- Azure AI Search indexes the documents
- Uses chunking + embeddings for semantic search
- (Optional) Extracts key fields like policy name, effective date, applicability, timelines

Runtime

User asks question → Agent queries Azure AI Search → retrieves top passages → LLM answers using only retrieved text → shows citations.

Sample Tests:

A) Policy Q&A (Grounded Answers)

Examples:

- “What is the weekly working hours limit?”
- “Who approves amendments to working hours guidelines?”
- “Is WFH allowed for all employees?”

B) Guided Processes (Step-by-step)

Examples:

- “How to raise a grievance in SPEAK UP?”
- “What happens after I submit a grievance?”
- “What’s the appeal timeline?”

C) Policy Boundaries + Disclaimers

Examples:

- If asked about salary disputes, performance appraisal, promotion etc.
The agent answers:
- “This is excluded from grievance scope and handled through a separate process.”

D) Drafts & Templates (Optional)

- Create a grievance statement template
- Draft an email to BU HR Head for exception cases
(Only if your organization allows it)

Q&A Section

Q1: Why do we need embeddings?

Because vector search understands meaning, not just keywords.

Q2: What happens if indexer fails?

Check error → Reset → Run again.

Q3: Why does Copilot Studio become ready instantly?

Because indexing already happened in Azure.

Q4: Can we update documents automatically?

Yes, schedule the indexer.

Q5: What’s the difference between index and indexer?

Indexer processes files.
Index stores searchable content.

Final One-Line Definition

Azure AI Search enables enterprise-level RAG implementations in Copilot Studio by allowing full control over indexing, embeddings, and semantic search.

Copilot Studio Standard Knowledge Vs Azure AI Search Knowledge

Feature	Copilot Studio (Standard Knowledge)	Azure AI Search (as Knowledge Source)
Complexity	Low-code: Point and click to add files or websites.	Pro-developer: Requires configuring indexes, vectorization, and rankers.
Search Tech	Traditional keyword search or basic semantic index.	Advanced Vector , Keyword , and Hybrid search with a Semantic Ranker .
Data Types	PDF, Word, SharePoint, and public websites.	Structured (SQL/NoSQL) and massive unstructured datasets (Blob storage).
Scalability	Best for smaller, departmental knowledge bases.	Built for high-performance , enterprise-scale data retrieval.
Customization	Limited control over how data is indexed or ranked.	Full control over query filters, indexing strategies, and RAG logic.

Key Differences & Use Cases

- **Direct Knowledge Sources (Copilot Studio):**
 - **When to use:** You need a quick solution for common business documents (e.g., HR handbooks, policy PDFs) or SharePoint sites.
 - **Strengths:** Rapid deployment and seamless integration with **Microsoft 365**.
 - **Limitations:** Performance may degrade with extremely large or complex datasets that require precise fine-tuning.
- **Azure AI Search (Integrated with Copilot Studio):**

- **When to use:** Your data is stored in complex formats like Azure SQL or large Blob storage, or you require "hybrid search" to improve response accuracy.
- **Strengths:** Supports **Vector Search** (semantic understanding) and **Integrated Vectorization**, allowing the agent to understand the "meaning" behind a user's query rather than just matching keywords.
- **Setup:** You must link your pre-built index using an API key and endpoint within the Copilot Studio Knowledge tab

What is Vector database?

A **vector database** is a specialized storage system designed to manage "vector embeddings"—numerical representations of data that capture its **meaning and context** rather than just its literal text.

While traditional databases (like SQL) are built for structured data in rows and columns, vector databases are purpose-built for the high-dimensional data used by AI models to perform **semantic search**.

How It Works

1. **Vectorization:** An AI model (like a Transformer) converts unstructured data—such as a PDF, image, or audio clip—into a "vector," which is a long list of numbers (e.g., [0.1, -0.5, 0.8...]).
2. **Mapping Meanings:** In this numerical space, similar concepts are placed physically closer together. For example, the vectors for "cat" and "kitten" would be mathematically near each other, while "cat" and "car" would be far apart.
3. **Similarity Search:** When you ask a question, the database doesn't look for matching keywords; it looks for the vectors **closest to your query's meaning**.

Why It's Essential for AI (RAG)

Vector databases are the "memory" for Large Language Models (LLMs) through a process called **Retrieval-Augmented Generation (RAG)**:

- **Preventing Hallucinations:** Instead of the AI "guessing" an answer, the vector database retrieves factual, private documents to provide as context.
- **Up-to-Date Info:** LLMs have a "cutoff date" for their knowledge. A vector database allows them to access your latest company data in real-time.
- **Handling Complexity:** They can search through millions of documents in milliseconds to find exactly what's relevant to a specific user prompt.

Traditional vs. Vector Databases

Feature	Traditional Database (SQL/NoSQL)	Vector Database
Data Format	Structured (Tables, JSON)	Unstructured (Embeddings)
Search Type	Exact Keyword Match	Semantic Similarity
Typical Query	"Find user with ID 123"	"Find documents about hiking in rain"

Use Case	Inventory, HR, Finance	AI Chatbots, Recommendations
----------	------------------------	------------------------------

Summary Comparison Table

Component	Role	Complexity	Best For
Copilot Studio	The "Face" (Frontend)	Low (No-code)	Quick deployment & UI
Azure AI Search	The "Brain" (Retrieval)	Medium (Pro-code)	Accuracy & Scale
Vector DB	The "Library" (Storage)	Technical	Understanding Context

CONFIDENTIAL